Medhansh Garg

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Education

University of Illinois, Urbana-Champaign Aug. 2023 – May 2027 (Expected)
- B.S. in Computer Engineering | Relevant Coursework: Data Structures & Algorithms, Computer Systems Engineering, Computer Architecture, Operating Systems, Databases

Skills

- **Languages**: TypeScript, Python, Java, C, C++, SQL, Kotlin, Go
- **Frameworks & Web**: React, Next.js, Node.js, FastAPI, Flask, Spring Boot, Prisma, SQLAlchemy, REST APIs
- **Data & Cloud**: PostgreSQL, SQLite, Supabase, Docker, Kubernetes, AWS, Azure, GitHub Actions CI/CD, nginx, Git, Linux
- **Systems**: RISC-V, QEMU, WebAssembly, Emscripten, Android HALs
- **ML & CV**: PyTorch, MediaPipe, OpenCV, YOLOv8, NumPy, pandas

Experience

Hopcharge, Gurugram, India | *Software Engineering Intern* Jun. 2026 – Aug. 2026
- Built the **Ad Engine**, a marketing pipeline (**Next.js 16**, **React 19**, **Prisma**/**Postgres**) that generates, reviews, and publishes ad creatives to **Meta** and **YouTube** through config-swappable vendors, with a feedback loop tracking CPL and engagement to guide new creatives. Scaled output from 2–3 ads/week to **3/day**, about **8**×
- Built **VOLT.CV**, a **FastAPI** HR dashboard automating hiring end-to-end: resume parsing with 3-tier OCR fallback (**pdfplumber**, **PyMuPDF**, **Tesseract**) and **spaCy** field extraction, Gmail reply classification, and bulk interview scheduling
- Secured it with database-level **Row-Level Security** tied to Google identities in **Supabase**/**PostgreSQL**, and shipped a cross-platform desktop build with **PyInstaller** and **GitHub Actions** CI, targeting 4 OS/architectures
- Built **Grid Ops**, replacing 5 hand-run scripts with a **FastAPI**/**React** (**Vite**/**TS**) service covering demand mapping, churn, forecasting, and pod allocation across **3 Delhi-NCR metros** (about 60 hexes), deployed behind **nginx** on an ARM VM

CS 124 Honors, Urbana, IL | *Lead Web Developer* Aug. 2025 – Present
- Lead development of the CS 124 platform (**Next.js**, **Supabase**), hosted on **Vercel** with **GitHub Actions** CI deploying directly to it, serving **300+ students** and **30+ staff** each semester. Built **CILogon** SSO and a 6-role least-privilege system with temporary role-view access, no permanent escalation
- Routed all public pages through a server-side API layer so no DB keys reach the client. Built a **246-test Vitest**/**Playwright** suite (6 unit, 167 API/DB integration, 73 browser E2E), with E2E gating dev-to-main promotions in **GitHub Actions** CI
- Automated student/staff, grading, and event management: batch grading, CSV import, and sprint tracking. Lead an **11-person** dev team through weekly meetings and PR reviews

Care Health Insurance, Gurugram, India | *Cybersecurity Intern* Jun. 2025 – Aug. 2025
- Ran external recon that uncovered **60+ unauthenticated internet-exposed services**, **30** of which were pulled internal after my findings; reverse-engineered **WAF request-signing** into a **Burp Suite** replay PoC, and bypassed **SSL pinning** and root detection (**Frida**, **JADX**) to unlock dynamic analysis of the customer mobile app

IoT++, Remote | *Software Engineering Intern* Jun. 2024 – Aug. 2024
- Built a real-time fire/smoke detection pipeline with **YOLOv8**, improving detection accuracy **62%** relative to the base YOLO model and throughput **13**× through **pruning** and **tiling**. Shipped **NPU**-optimized inference on **12 Orange Pi units** in factory pilot, containerized with **Docker** and **Kubernetes** on **Azure**

Projects

ECE 391 RISC-V Kernel | *C, RISC-V, QEMU* github.com/HackOverflow404/ece391-kernel-demo
- With a team, wrote a **RISC-V64 kernel** from scratch: **VirtIO** console and block drivers, **KTFS** filesystem, page-table virtual memory, scheduler with kernel threads, user/kernel syscalls, and a Unix-style shell with multi-stage pipelines
- Compiled **TinyEMU** to **WebAssembly** (Emscripten) so the kernel boots live in the browser, with an **xterm.js** shell and filesystem mounted over HTTP: ece-391-kernel-demo.vercel.app

Echo Show 5: LineageOS & Camera Pipeline | *C++, Kotlin, HALs* github.com/HackOverflow404/EchoCameraStreamer
- Ported **LineageOS 18.1** via the **amonet bootrom exploit**; fixed a **MediaTek camera ABI/vtable mismatch** and a defconfig bug
- Built a **Kotlin 720p H.264** webcam app and **SECUR-T**, a Raspberry Pi hub streaming to go2rtc/RTSP and Home Assistant; tuned **High profile** encoding to cut blocky artifacts **38%** on the same scene
- Added luma-only **denoise + sharpen** (∼1/3 core; optional **RTX 4070** path); on-demand pipeline cut idle CPU **12–14%** to **0.1%**

AlberFlowy | *Node.js, C++, OAuth2* github.com/HackOverflow404/AlberFlowy
- Reverse-engineered **WorkFlowy's private API** for full tree CRUD through a **Node.js** CLI and **C++ Albert plugin**; daily-driven on my 150+ node vault, with cached live queries at **0.7ms p50** / **1.7ms p95**

ASL Adaptive Tutor | *PyTorch, MediaPipe* github.com/HackOverflow404/SignLanguageLearning
- Built a real-time ASL tutor with a **PyTorch triplet-embedding** grader over **MediaPipe** pose landmarks: **85.2% top-1** / **97.8% top-5** across **60 signs**, **2.67**× a DTW baseline

Leadership & Involvement

Illinois MicroTech, Urbana, IL | *External Vice President* | *EOH Project Lead* Oct. 2025 – Present
- Led a **15-person** team delivering the club's Engineering Open House 2026 showcase to **20,000+ attendees** on a **$250** budget
- Manage the club's marketing chairs, EOH leads, and outreach committee, alongside independent outreach initiatives
- HackerFab: engineering a **3-axis stepper-driven stage** with **sub-nm positioning** for wafer photolithography